

COURSE TITLE : MAINTENANCE OF MACHINES
COURSE CODE : 5301
COURSE CATEGORY : A
PERIODS / WEEK : 4
PERIODS / SEMESTER : 60
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Objectives of Plant Maintenance Types Of Maintenance	15
2	Preventive Maintenance Organization Of Maintenance Department	14
3	Lubrication & Spare Parts Network Systems In Maintenance	17
4	Replacement Analysis Maintenance Tools	14
TOTAL		60

COURSE OUTCOME:

1. Explain the Objectives of Plant Maintenance.
2. Understand different Types of Maintenance.
3. Understand the Concept Preventive Maintenance.
4. Explain the Organisation Of Maintenance Department.
5. Understand about Lubrication & Spare Parts.
6. Understand Network Systems in Maintenance.
7. Explain about Maintenance Tools.
8. Describe Replacement Analysis.

MODULE I

1.1.0 Explain the Objectives of Plant Maintenance.

- 1.1.1 Enlist the Objectives of Plant Maintenance.

1.2.0 Understand different Types of Maintenance.

- 1.2.1 Explain Corrective or break down maintenance.
- 1.2.2 Explain Scheduled maintenance.
- 1.2.3 Explain Preventive maintenance.
- 1.2.4 Explain Break down maintenance.

MODULE II

2.1.0 Understand the Concept Preventive Maintenance.

- 2.1.1 Describe the Concept and Repair cycle.
- 2.1.2 Explain the Classification of production equipments.
- 2.1.3 Understand Annual and monthly preventive maintenance schedule.
- 2.1.4 Know the importance of Preventive maintenance records.
- 2.1.5 List out the Advantages of preventive maintenance.

2.2.0 Explain the Organisation Of Maintenance Department.

- 2.2.1 Explain Organization structure of maintenance department
- 2.2.2 Understand Maintenance centre for each shop and their Duties & functions.
- 2.2.3 Understand the functions of various sections of maintenance workshop.

MODULE III

3.1.0 Understand about Lubrication & Spare Parts.

- 3.1.1 Explain the Purpose of lubricants.
- 3.1.2 Understand Replacement of lubrication schedule.
- 3.1.3 Understand Replacement of spare parts.
- 3.1.4 Explain Bearings, belt and their specifications.
- 3.1.5 Understand Simple methods of repair.

3.2.0 Understand Network Systems in Maintenance.

- 3.2.1 Enlist the advantages of using PERT & CPM in maintenance.

MODULE IV

4.1.0 Describe Replacement Analysis.

- 4.1.1 Understand Replacement Analysis and Reasons for replacement.
- 4.1.2 Explain the factors considering for replacement.

4.2.0 Explain about Maintenance Tools.

- 4.2.1 Understand the Common tools required for maintenance.

CONTENT DETAILS

MODULE I

Objectives of Plant Maintenance-Types of Maintenance-Corrective or break down maintenance, Scheduled maintenance, Preventive maintenance, Break-down maintenance.

MODULE II

Preventive Maintenance-Concept -Repair cycle-Classification of production equipments-Annual and monthly preventive maintenance schedule-Preventive maintenance records-Advantages of preventive maintenance-Organization of Maintenance Department-Organization structure of maintenance department-Maintenance centre for each shop-Duties & functions-Functions of various sections of maintenance workshop.

MODULE III

Lubrication & Spare Parts-Purpose of lubricants-Replacement of lubrication schedule-Replacement of spare parts-Bearings, belt-their specifications-Simple methods of repair- Network Systems in Maintenance-Advantages of using PERT & CPM in maintenance.

MODULE IV

Replacement Analysis-Introduction-Reasons for replacement-Factors consider for replacement-Maintenance Tools-Common tools required for maintenance.

REFERENCES

1. Industrial Maintenance by S Garg
2. Industrial Engineering & Management by O.P Khanna